Christos Giannakopoulos

Analytic Scientist & Computational Physicist
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Personal Webpage

PROFESSIONAL SUMMARY

Data Scientist and Computational Physicist with a Physics Ph.D. and **5+ years** of hands-on **predictive modeling, statistical analysis, and data mining** on large-scale, multi-modal datasets. Deep expertise in Python (5+ years) and MATLAB (6+ years), with a proven track record applying Gaussian process regression, Bayesian inference, neural networks, and dimensionality reduction to complex real-world data. Experienced leading cross-functional teams, performing production data validation, and communicating quantitative findings to non-technical stakeholders under strict delivery deadlines.

TECHNICAL SKILLS

Programming: Python (5+ yrs), MATLAB (6+ yrs), SQL (2+ yrs), Bash
Statistical Modeling: Gaussian processes, Bayesian inference, non-linear regression, logistic regression, PCA/SVD, dimensionality reduction, feature engineering
Machine Learning: scikit-learn (logistic regression, random forests, SVMs, decision trees), TensorFlow, Keras, neural networks (CNNs), anomaly detection
Data Engineering: ETL pipeline design & monitoring, HPC/SLURM distributed computing, multi-terabyte data processing.
NLP & AI: spaCy, BERTopic/KeyBERT, LLM fine-tuning, Agentic RAG
Dev Tools: Git/GitHub, Docker, Linux

EXPERIENCE

Research Scientist — *University of Cincinnati* 2021 – 2026

Research Scientist embedded in a multi-institution, federally funded program. Owned analytics and modeling infrastructure for a 50+ person collaboration with deliverables reviewed by leadership.

- Built and optimized **predictive models** on multi-modal datasets (time series, spatial maps) using Gaussian process regression, Bayesian inference, and non-linear optimization, improving signal detection sensitivity by **2×.**
- Conducted extensive **data mining via feature engineering and dimensionality reduction** (PCA/SVD) on high-dimensional sensor datasets, analyzing model behavior and extracting insights that improved model confidence by **95%.**
- Built **production data validation frameworks** that cross-validated model outputs against independent measurements across all stages of acquisition including normalization, and transformation, which enabled data-driven decisions for program stakeholders and meeting analysis deadlines **50% sooner**.
- Developed **model validation frameworks** to quantify previously unmeasured systematic uncertainties and benchmark end-to-end system performance, informing hardware design trade-offs and achieving a **78% reduction in compute costs**.
- Designed, implemented, and monitored **production ETL pipelines** processing **multi-terabyte datasets** on distributed HPC/SLURM clusters, ensuring data quality through all stages of collection, normalization, and transformation. Outputs were version-controlled and adopted as the **program-wide analytical standard** for a 50+ person multi-institution collaboration.
- **Redesigned core pipeline architecture**, improving runtime by **8×** and enabling near real-time data quality monitoring across distributed compute infrastructure.
- Led a **cross-functional team** on a six-month analytical project, delivering validated outputs reviewed and approved by external program leadership as the collaboration's analytical standard.
- Built an **end-to-end NLP data pipeline** using spaCy to ingest and process large-scale, multi-source unstructured data, enabling named entity recognition and topic detection, reducing manual review time by **35%.**
- Authored **publications** establishing the mathematical and computational framework for novel analysis adopted across the program.

EDUCATION

Ph.D. in Physics — University of Cincinnati 2026

B.S. in Physics — Hillsdale College 2019

CERTIFICATIONS & TRAINING

Machine Learning Specialization — DeepLearning.AI
3-course specialization covering Neural Networks, CNNs, Random Forests, Reinforcement Learning, Anomaly Detection, Clustering, and Recommender Systems.

Experimental Systems Engineering — Harvard University & Caltech
Designed and characterized precision measurement systems; performed end-to-end hardware validation and statistical error analysis for a next-generation sensor array.

AWARDS & PUBLICATIONS

"**Calibration Measurements of the BICEP3 and BICEP Array CMB Polarimeters from 2017 to 2024**" Christos Giannakopoulos — SPIE — 2024 *(predictive modeling, data analysis, statistical analysis, data mining)*

Best Recitation Instructor — Exceptional Teaching Award University of Cincinnati *2019, 2020*